

ECE6370 ADVANCED DIGITAL DESIGN

Lab 4: ROM-based Game Access Control on FPGA

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Description of the System:

A mental binary math game that can be played by a single person on the FPGA board. First to start the game we need to press reset button and then the system will authenticate the player by using an Access Controller module. The Access Control module will authenticate the players by storing the default password in a ROM. Use my last four digits of PeopleSoft ID as a default password. Note that only the game creator's ID will be checked. In other words, I am giving permissions to the players to play my game. Separating the timer control function from the Password Authentication function and replacing the 1ms timer module with a more cost-efficient one based on Linear Feedback Shift Register (LFSR). After each digit is set on toggle switches (4 bits in its binary form), the push button is pushed to enter that digit. After all 4 digits are successfully entered and verified, the player is allowed to play the game and to enter the password digits use password toggle switches. If entered password is correct, then right most LED will light up and if entered password is incorrect, then right second most LED will light up. Once we logged in to the system we can start playing the game. The game is controlled by a timer (default to 99 seconds). The player will try to complete as many rounds as possible before timeout. Remaining game time (in seconds) should be displayed in base 10. Before the password is successfully verified, the timer should display "00". When the password is checked, 99 seconds will be loaded to the timer. Pushing the GAME Button once will start the game timer. After the timer reaches zero, the player is not allowed to enter any more matching numbers or generate any new random numbers. When the game is over, pushing the GAME Button again will reload the initial timer value ("99" seconds). Pushing the same button again will start the timer. A hardware-based true Random Number Generator is developed to generate a number between 0 and 14. To generate a random number, the Player will push a button for an unspecified amount of time and release the button afterwards. This button is not be shaped by the Button Shaper as the duration of the button push is used to trigger the LFSR counter to change value for a random period of time (When the button is pushed, a 4-bit LFSR counter will changes at the rate of 50 MHz until the button is released where the last number on the LFSR counter is displayed for the Player to match). Before passing the password check, pushing the random number generation button or the Player's load button will not cause a new random number to be generated. We achieved this easily by sending the random number generating button signal to the access controller. The access controller will force its output button signal to be 1 as if the button is never pressed before the password is verified. Once the password is verified, the button output signal from the access controller should follow the input button signal. We should not invert the button signal inside the access controller. We will do it inside the Random Number Generator module (to trigger a 4-bit LFSR counter to count).

The player will set a 4-bit binary number from his/her set of toggle switches and then push his/her button to load the number into the system and the number will be displayed on one of the 7 segment display devices. He or she should mentally convert the Random number observed into the binary version and then identify the matching number so that the sum of the two numbers is 1111 in binary. The sum of the two numbers will be computed by the FPGA and displayed on a third 7-segment display device. In all, we need five 7-segment display devices from the board to display Player's number, Random number, Sum and the Timer, respectively. The player can play multiple times during the timer running from 99 to 00, once the timer is stopped the player is not allowed to play, he or she need to restart the game again by pressing the game start button. Before the load button is pushed, any intermediate changes to the toggle switch settings will not be displayed or used by the system for computation. The number entered by Player will be loaded to the system if the corresponding

LOAD button is pushed. Before the game is authenticated, Player cannot load the number even if the LOAD button is pushed. On top of it, a verification module is added to automatically check if the sum reaches 1111, and if so, light up the left second most LED(Matching), and if not, light up the left most LED(Non-Matching).

Logout function: After the user is logged in, when the game is not in an active session (the timer is not running), the user can press the second button from the left (Player's Load button) to log out. After the user is logged out, all game functions including the timer will be disabled.

LFSR-based counter (in the Random Number Generator): Replaced the 4-bit counter inside the Random Number Generator with a more cost efficient 4-bit LFSR counter. Therefore it generates values in the range 0 to 14 by excluding 15(F).

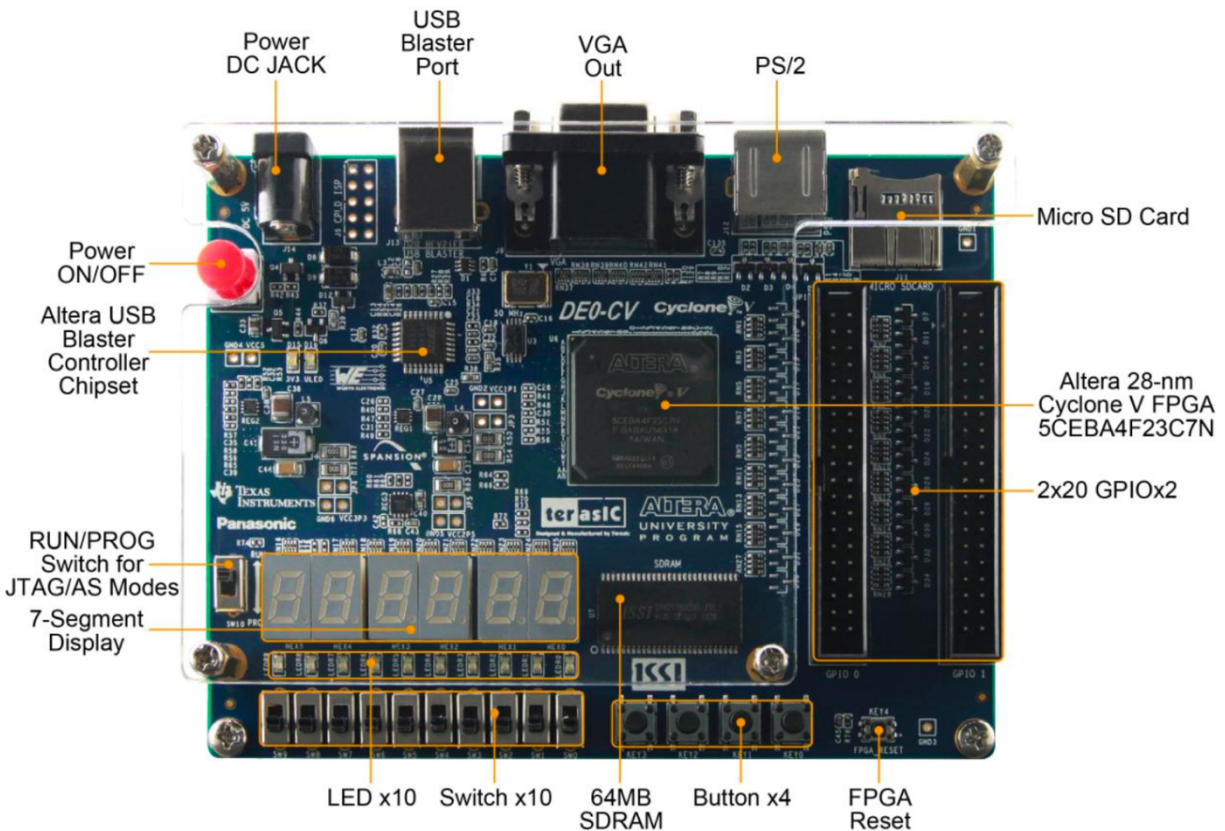
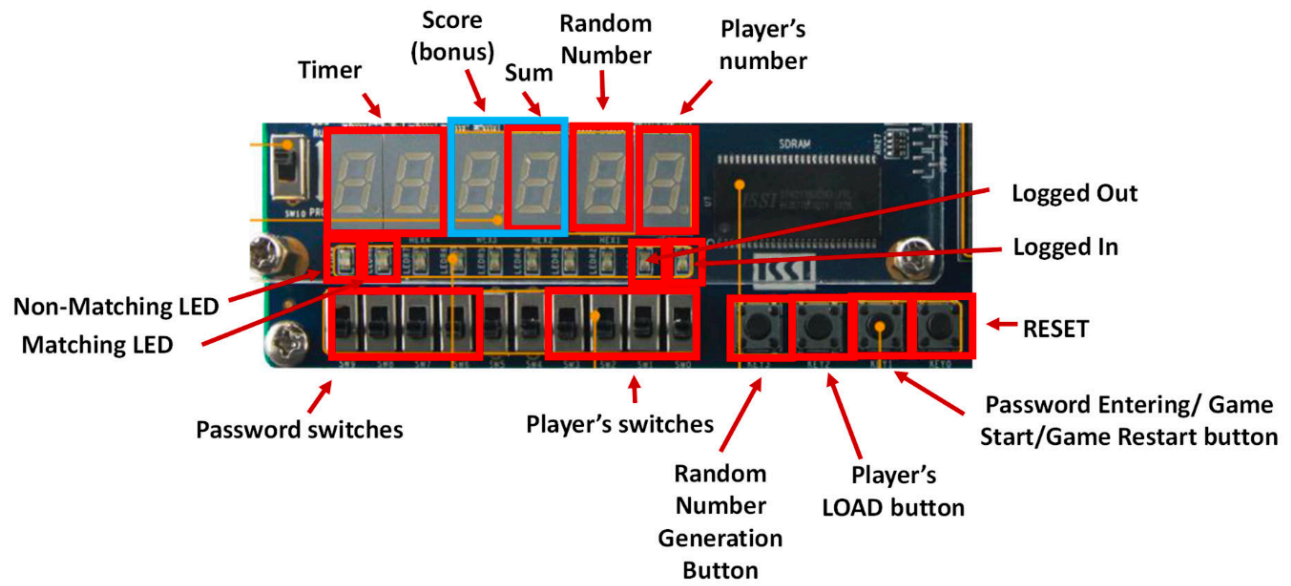


Figure 1-2 Development Board (top view)



How to Play:

1. First, make sure all the switches are OFF indicating binary 0, FPGA is in RUN mode and Power is ON.
2. Initially all the above highlighted 7-segment displays will display 0, Logged Out LED will be glowing because we are not yet authenticated, and Non-matching LED will be glowing because the Sum is 0 and Matching LED will not glow because the Sum is not 15(F).
3. First to start the game press RESET button which will start the system with default values and then the system will authenticate the players by entering password.
4. Use my last four digits of PeopleSoft ID as a default password(4090). Note that only the game creator's ID will be checked. In other words, I am giving permission to the player to play my game.
5. To enter the password digits, use password toggle switches. After each digit is set on toggle switches (4 bits in its binary form), the password entering button must be pushed to enter that digit. Likewise, we need to push password entering button for every digit of password.
6. Meanwhile, if player enters a number by his/her toggle switches and push his/her LOAD button, then the system will not accept that number and still 0 is displayed. Before the game is authenticated, Player cannot load the number even if the LOAD button is pushed.
7. Similarly, Player cannot generate a random number by pressing the Random Number Generation Button before the game is authenticated.

8. After all the 4 digits are successfully entered and verified, the player is allowed to play the game. If entered password is correct(4090), then right most LED(Logged In) will light up and if entered password is incorrect(Ex. 4098), then right second most LED(Logged Out) will light up.
9. Once we logged in to the system the Timer is set to 99. To start the game, we need to press the game start button. Before player either cannot generate a random number or enter a number.
10. Once the game start button is pushed, the player can start playing the game and also Timer start counting down the seconds from 99 to 00. In the First Round, Player should first generate a Random number by pressing random number generation button, then player can see the random number on the display. For instance, Random number displaying is 10.
11. Player should set a number using his/her toggle switches in binary form and must push Player's LOAD button to load the number into the system, then the number is displayed on the Player's display. Before the load button is pushed, any intermediate changes to the toggle switch settings will not be displayed or used by the system for computation.
12. Player should be careful while toggling switch because he/she should set a number such that the sum of the player's number and Random Number displaying must be equal to 1111. For instance, Player wants to set number 4, then he/she need to toggle switches by making them 0100 and push Player's LOAD button.
13. Therefore, the Sum is 14→E it is not equal to 15, the Non-matching LED will be still glowing which indicates Player set a wrong number so he/she loss this match/round.
14. Before starting the next round, you must toggle all the switches back to 0.
15. In the Second Round, Player should press the random number generation button as long as you wish, once the button is released a random number is displayed on the Random number display. For instance, Random number displaying/generated is 12.
16. Now the Player should set a number using his/her toggle switches in binary form and must push Player's LOAD button to load the number into the system, which will be displayed on the Player's display. Player should be careful while toggling switch because he/she should set a number such that the sum of both the player's number and Random Number displaying must be equal to 1111. For instance, Player wants to set number 3, then he/she need to toggle switches by making them 0011 and push Player's LOAD button.
17. Therefore, the Sum is 15→F it is equal to 15, the Non-matching LED will be OFF/ not glows and matching LED will start glowing which indicates Player set a correct number, so he/she wins this match/round.
18. By this way the player can play multiple rounds and note the score to evaluate total losses/wins.
19. Toggle switches order is from left to right while setting a number that is toggle switches follows the binary number order (Example: SW3 SW2 SW1 SW0 → 1 0 1 0 {10→A}).

20. Once the Timer shows 00, it indicates timeout, and player cannot play the game. If he/she wants to enter any number by pressing the random number generation button will not display any new value. Note the player is still logged in to the game.
21. To Restart the game, player should push the game start button. Once the button is pushed the Timer is reconfigured to 99, but game is not yet started any load button press will not display any new number.(Note: password entering, game start and game restart functions uses the same button.)
22. Player should press the game restart button again to start the game. Once the button is pressed the Timer starts counting down from 99 to 00 and player can start playing the game again.
23. At the end of the game when the timer is 00 and game is in inactive mode if player wants to log out from the game, then by pressing the player's LOAD button will Log Out from the game. Now, Logged Out LED will starts glowing and Logged In LED will stops glowing. To start the game player must enter a valid password to log in to the system. (Note: player's LOAD button is also used as Log Out button only when timer is 00 and game is inactive).
24. During the Game or end of the game if player want to take break or stop from playing, then pressing the RESET button will Log Out from the system and Now, Logged Out LED will starts glowing and Logged In LED will stops glowing. This will flush all the values stored in the game.
25. After the system is logged out, pushing the LOAD button or random number generation button will not load any new number. The player will be allowed to log in again to start the new game.
26. Player can refresh/restart game by pressing the RESET button which will erase previous records and assign default values(0) to the displays. A correct password must be entered to Log In into the system. If player wants to start the game, he/she is allowed to enter password again to Log In into the system.